

- Two spheres have equations $|\vec{s} - \vec{c}_1| = 3$ and $|\vec{s} - \vec{c}_2| = 4$ where $\vec{c}_1 = (1, 5, 3)$ and $\vec{c}_2 = (-1, 2, 9)$
 - Show that the two spheres touch each other.
 - Find the intersection point P of the two spheres.
 - Find the equation of the plane which tangent to both spheres and include point P .
- Two spheres have equations $|\vec{s} - \vec{c}_1| = 5$ and $(x - 3)^2 + (y + 1)^2 + (z + 3)^2 = 24$ where $\vec{c}_1 = (1, 2, 3)$
 - Show that two spheres intersect each other.
 - Find the center and the radius of the circle of intersection.
- Let n be a positive integer.

a) Show that $\frac{1}{1-t^2} = (1 + t^2 + t^4 + \dots + t^{2n-2}) + \frac{t^{2n}}{1-t^2}$ for $t^2 \neq 1$

b) For $-1 < x < 1$, show that

$$\int_0^x \frac{t}{1-t^2} dt = \ln\left(\frac{1}{\sqrt{1-x^2}}\right)$$

c) Using the above parts and by letting $x = \sqrt{\frac{8}{9}}$, deduce that

$$\int_0^{\sqrt{\frac{8}{9}}} \frac{t^{2n+1}}{1-t^2} dt = \ln 3 - \sum_{k=1}^n \frac{1}{2k} \left(\frac{8}{9}\right)^k$$

d) If $0 \leq t \leq \sqrt{\frac{8}{9}}$, show that

$$0 \leq \ln 3 - \sum_{k=1}^n \frac{1}{2k} \left(\frac{8}{9}\right)^k \leq \frac{9}{2n+2} \left(\frac{8}{9}\right)^{n+1}$$

- For real numbers $x_1, x_2, \dots$. Define

$$S_n = \sum_{k=1}^n x_k$$

a) Show that

$$\sum_{k=1}^n x_k^2 = \frac{1}{n} S_n^2 + \sum_{k=1}^n \left(x_k - \frac{1}{n} S_n\right)^2$$

b) Hence show that for $n = 2, 3, 4, \dots$,

$$\sum_{k=0}^n \binom{n}{k}^2 > \frac{2^{2n}}{n+1}$$

c) Use the identity $[(1+x)^n]^2 \equiv (1+x)^{2n}$ to show

$$\sum_{k=0}^n \binom{n}{k}^2 = \frac{(2n)!}{(n!)^2}$$

d) Hence or otherwise show that $0 < \ln 2 - \ln 3 + \ln 4 - \dots + \ln(2n) < \ln(n+1)$ for $n = 2, 3, 4, \dots$

5. For $2 \leq k \leq n$

a) Show that

$$\frac{\binom{n}{k}}{n^k} = \frac{\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{k-1}{n}\right)}{k!}$$

b) Deduce that

$$\frac{\binom{n+1}{k}}{(n+1)^k} > \frac{\binom{n}{k}}{n^k}$$

c) Deduce that, if n is a positive integer,

$$\left(1 + \frac{1}{n+1}\right)^{n+1} > \left(1 + \frac{1}{n}\right)^n$$

6. n is positive integer and $z^2 \neq 1$

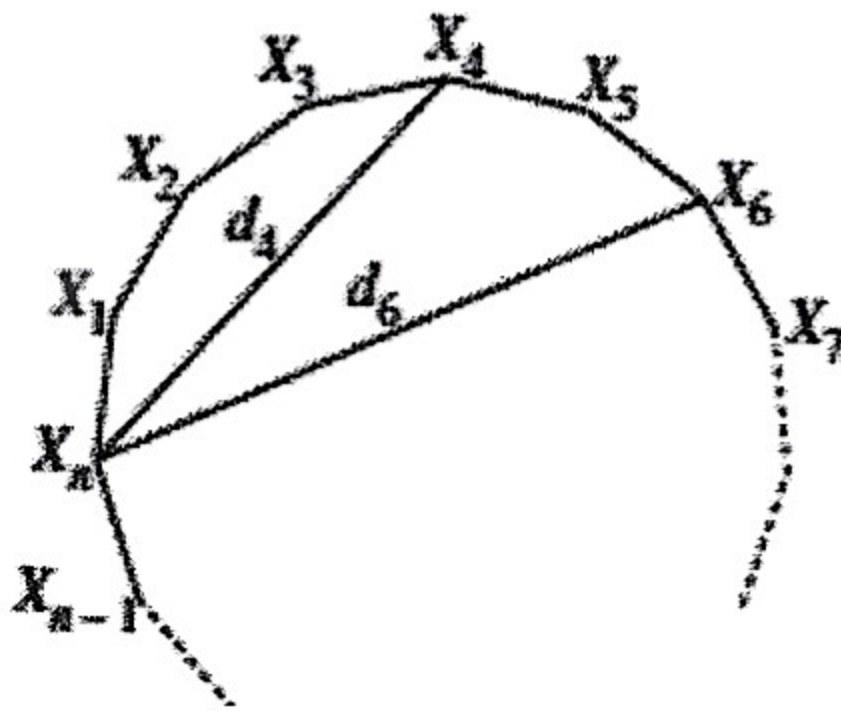
a) Show that

$$1 + z^2 + z^4 + \dots + z^{2n-2} = \left(\frac{z^n - z^{-n}}{z - z^{-1}}\right) z^{n-1}$$

b) If $z = \text{cis } \theta$ and $\sin \theta \neq 0$, show that

$$1 + \cos 2\theta + \dots + \cos(2n-2)\theta + i[\sin 2\theta + \dots + \sin(2n-2)\theta] = \frac{\sin n\theta}{\sin \theta} [\cos(n-1)\theta + i\sin(n-1)\theta]$$

c) Suppose $\theta = \frac{\pi}{2n}$, show that $\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \dots + \sin \frac{(n-1)\pi}{n} = \cot \frac{\pi}{2n}$



d) The diagram shows a regular n -sides polygon with vertices $X_1 X_2, \dots, X_n$. Each side has unit length. The length d_k of the diagonal $X_n X_k$ where $k = 1, 2, \dots, n-1$ is given by $d_k = \frac{\sin \frac{k\pi}{n}}{\sin \frac{\pi}{n}}$.

(Do NOT prove this). Show that $d_1 + d_2 + \dots + d_k = \left(2\sin^2 \frac{\pi}{2n}\right)^{-1}$

e) Let p be the perimeter of the polygon and $q = \frac{1}{n}(d_1 + d_2 + \dots + d_{n-1})$. Show that

$$\frac{p}{q} = 2 \left(n \sin \frac{\pi}{2n}\right)^2$$

f) Hence calculate the limiting value of $\frac{p}{q}$ as $n \rightarrow \infty$